A Spatial Analysis of Unemployment in Croydon, London.

A) Executive Summary

This study identifies spatial inequalities in unemployment across Croydon, with statistically significant clustering at the Lower Super Output Area (LSOA) level rather than even borough-wide distribution. High–High unemployment clusters are concentrated in northern Croydon, while extensive Low–Low clusters dominate the south, confirming strong local spatial dependence. Unemployment is closely associated with neighbourhood deprivation; however, geographically weighted regression reveals it as a stronger predictor of unemployment in eastern and southern LSOAs, with weaker explanatory power elsewhere. These findings suggest that uniform employment interventions risk overlooking areas of greatest need (OECD,2019) instead spatially targeted policies are likely to be more effective.

B) Report: Contextualising Unemployment Using the Index of Multiple Deprivation

While unemployment rates provide a direct measure of labour market exclusion, they do not capture the broader socio-economic context in which unemployment occurs. The Index of Multiple Deprivation (IMD) is therefore introduced in this analysis to contextualise unemployment within wider patterns of deprivation at the neighbourhood scale.

Distribution of Unemployment Rates

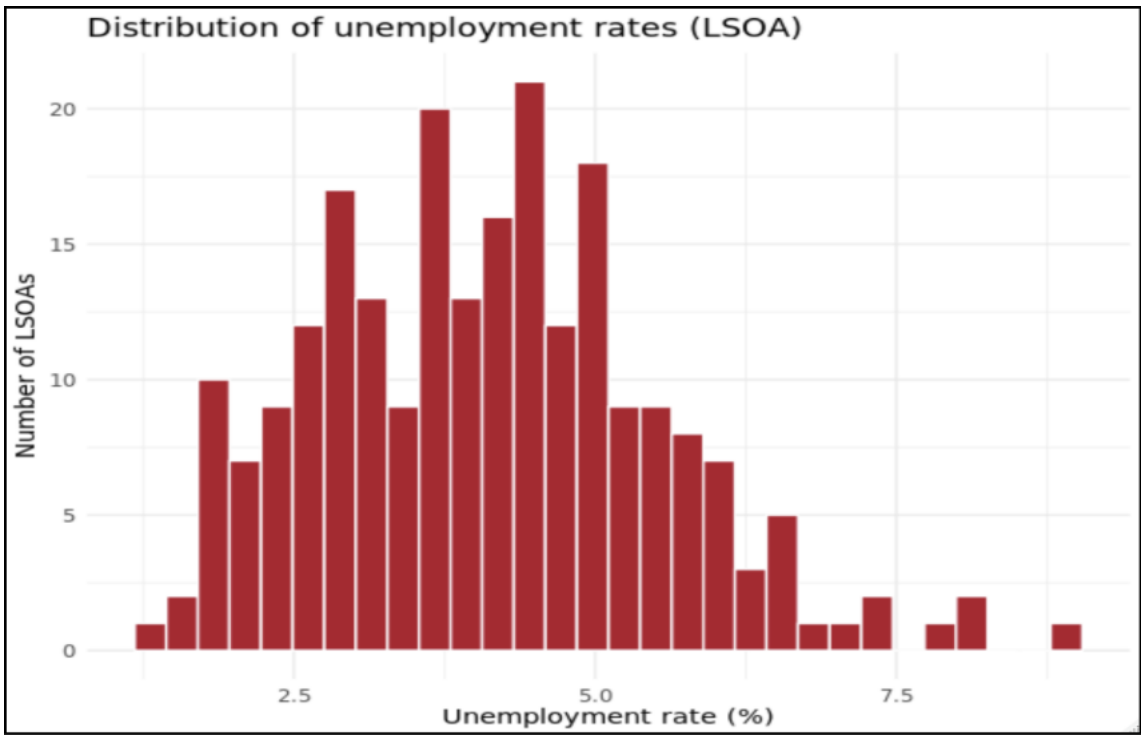

Fig.1 distribution of unemployment

The histogram shows moderately right-skewed distribution across Croydon LSOAs, with majority exhibiting unemployment rates between 2% and 5%, with smaller subset of areas experiencing higher rates. The mean unemployment rate is 4.11% (SD = 1.40), indicating notable variation between neighbourhoods (Overman & Puga,2010).

This distribution highlights a group of LSOAs with disproportionately high unemployment, suggesting that labour market disadvantage is unevenly distributed rather than randomly dispersed across the borough. This dispersion justifies a spatially explicit analysis motivating further examination of whether high-unemployment areas are geographically clustered.

Spatial Distribution of Unemployment

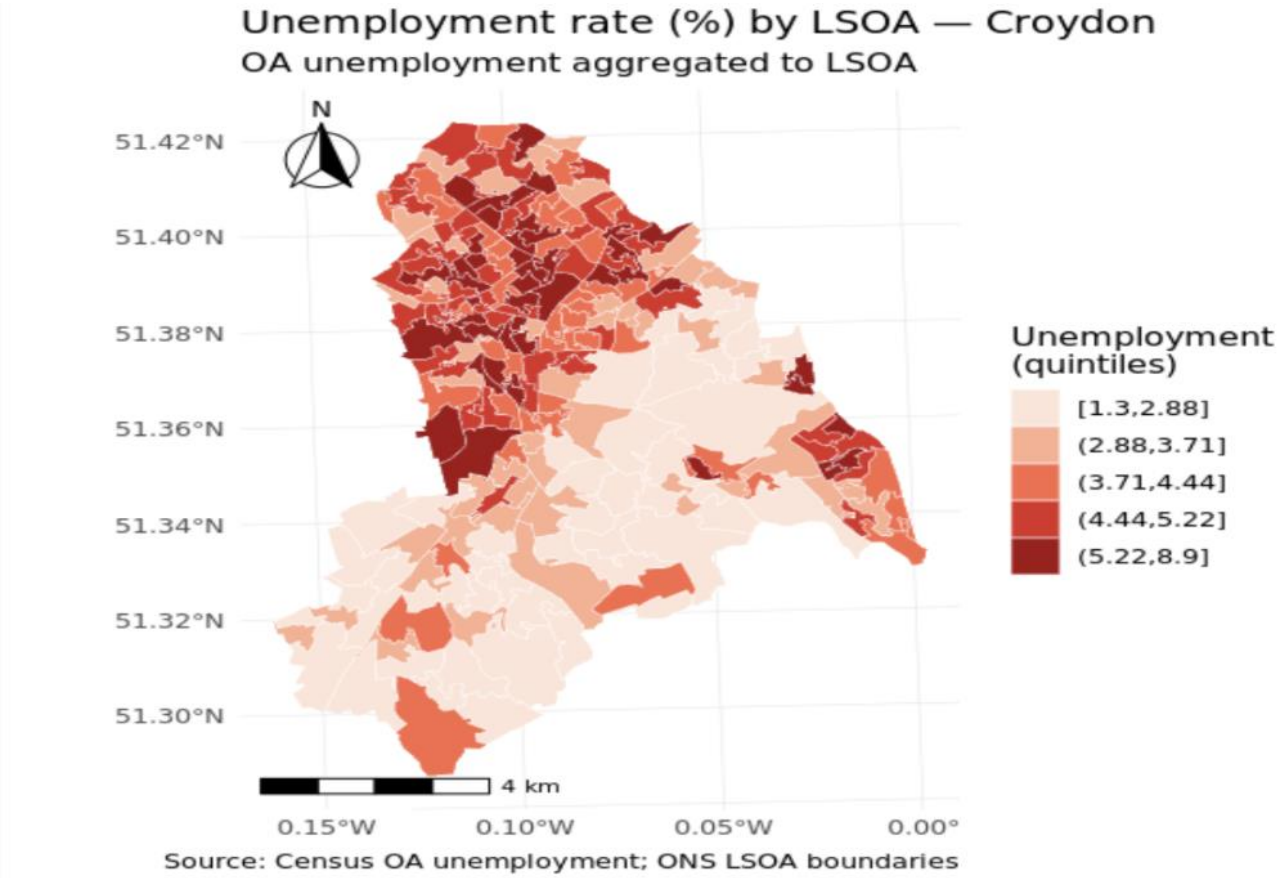

Fig.2 Unemployment rate(%) by LSOA

Figure 2 reveals clear spatial inequalities within Croydon. Higher unemployment rates are concentrated primarily in the northern and central parts of the borough, while much of the southern area exhibits comparatively lower rates. This divide is visually striking and aligns with known socio-economic contrasts (Martin,1997) and motivates formal spatial autocorrelation analysis.

Local Spatial Clustering of Unemployment (LISA)

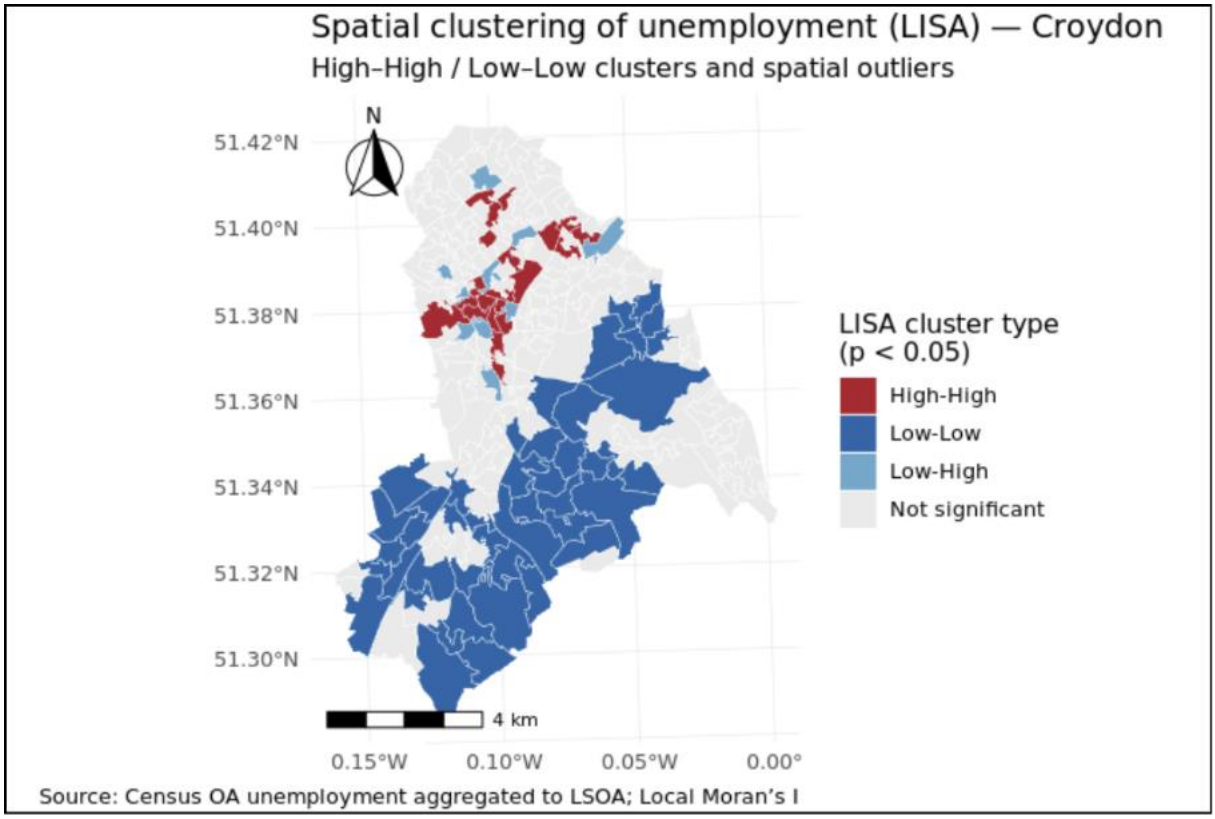

Fig.3 Local Moran I's of spatial clustering

Figure 3's Local Moran's I (LISA) analysis of unemployment rates at the LSOA level in Croydon shows statistically significant High–High clusters of unemployment are concentrated in northern parts while extensive Low–Low clusters dominate the south, reflecting spatial concentrations of low unemployment.

Several Low–High spatial outliers also exist, where relatively low-unemployment LSOAs border higher unemployment neighbours, highlighting sharp localised contrasts not captured by borough-wide averages.

Overall, the LISA results confirm strong local spatial dependence in unemployment demonstrating that aggregate measures mask substantial neighbourhood-level inequalities which provides clear justification for geographically targeted employment interventions rather than uniform borough-wide approaches.

Deprivation and Unemployment

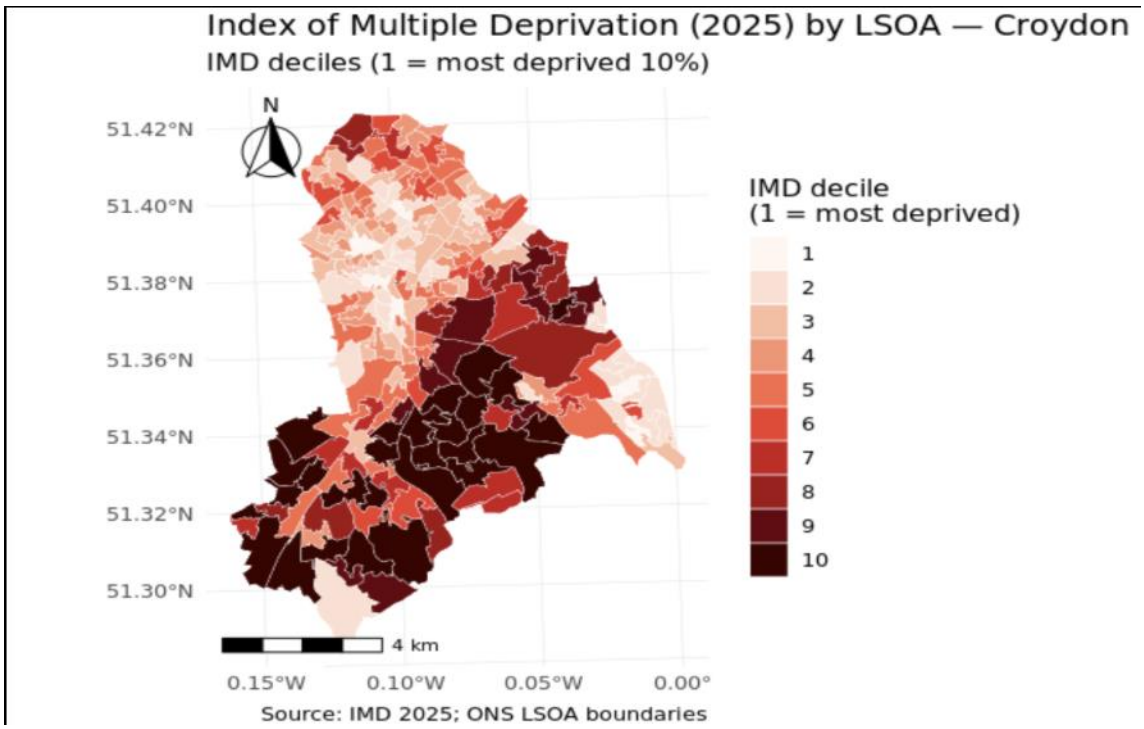

Fig.4 Croydon IMD decile by LSOA

Figure 4 illustrates the spatial distribution of deprivation using IMD deciles, with higher deprivation concentrated in many of the same northern and central LSOAs identified as unemployment hotspots (Mclennan et al., 2019). This visual map suggests a close relationship between deprivation and unemployment but does not indicate whether this relationship is consistent across space.

Spatial Variation in the Deprivation-Unemployment Relationship (GWR)

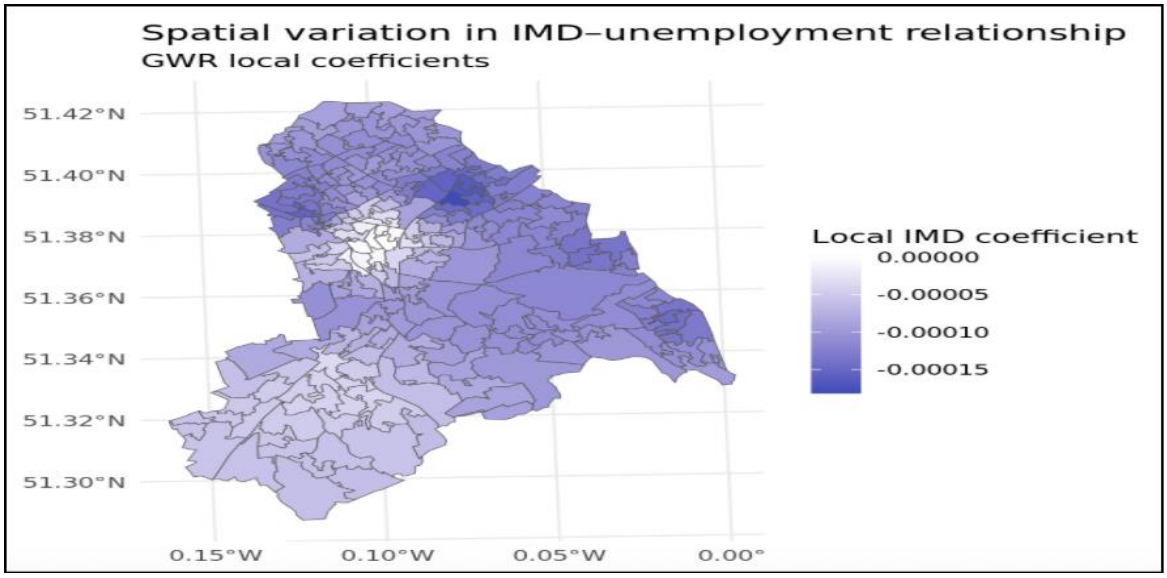

Fig.5 Spatial variation in unemployment deprivation relationship.

To analyse whether the strength of the relationship between unemployment and deprivation varies spatially, a geographically weighted regression (GWR) was estimated using IMD rank as the explanatory variable (X). Figure 5 maps the local GWR coefficients, revealing substantial spatial difference between the deprivation–unemployment relationship.

In northern and central Croydon, deprivation is a much stronger predictor of unemployment, indicating that socio-economic disadvantage moreso inhibits labour market outcomes in these neighbourhoods. In contrast, weaker coefficients in other areas suggest that unemployment may be influenced by omitted factors, such as commuting accessibility or demographic composition.

Local Coefficients and Model Fit

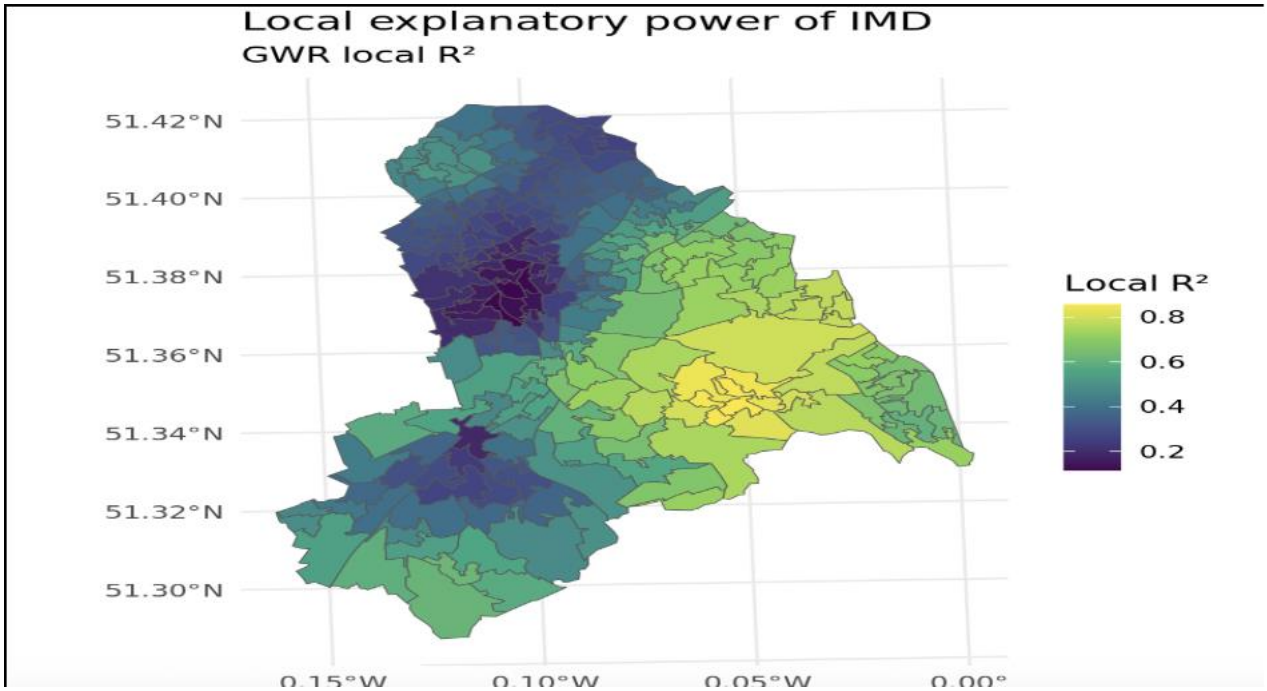

Fig.6 Local explanatory power of IMD; GWR local R^2

To assess local model performance, local R^2 values were extracted and mapped in figure 5 above. High local R^2 values (exceeding 0.6 in several eastern and southern LSOAs) indicate areas where IMD strongly explains unemployment variation, while substantially lower values in the north-west suggest the influence of alternative drivers.

In conclusion, these findings imply that borough-wide employment interventions risk overlooking neighbourhoods where deprivation most strongly constrains labour market participation (OECD) and suggestions include a focus on spatially targeted employment strategies focused on the most affected LSOA's.

Report references;

Martin, R. (1997) *Regional unemployment disparities and their dynamics*. Regional Studies, 31(3), pp. 237–252. https://doi.org/10.1080/00343409750134663

McLennan, D., Noble, S., Noble, M., Plunkett, E., Wright, G. and Gutacker, N. (2019) *The English Indices of Deprivation 2019: Technical Report*. London: Ministry of Housing, Communities and Local Government: https://www.gov.uk/government/statistics/english-indices-of-deprivation-2019

OECD (2019) *Local economic and employment development (LEED): Making decentralisation work*. Paris: OECD Publishing : https://doi.org/10.1787/g2g9faa7-en

Overman, H.G. and Puga, D. (2010) *Labour pooling as a source of agglomeration: An empirical investigation*. In: Glaeser, E.L. (ed.) *Agglomeration Economics*. Chicago: University of Chicago Press, pp. 133–150.

B) Technical Appendix: Rationale for methodology use

Data Aggregation and Spatial Units

Unemployment data were sourced at the Output Area (OA) level from the 2021 Census and aggregated to Lower Layer Super Output Areas (LSOAs) to enable integration with the Index of Multiple Deprivation (IMD), which is published at the LSOA scale. Aggregation was performed using population-weighted means to preserve relative unemployment intensity across spatial units (ONS,2021). This approach ensures consistency across datasets while retaining sufficient spatial resolution for neighbourhood-level analysis.

Jenks Natural Breaks

All choropleth maps were classified using the Jenks natural breaks method. This approach minimises within-class variance while maximising between-class variance (GISGeography,2023) allowing class boundaries to adapt to the empirical distribution of the data (Fotheringham et al.,2002).

Given the positively skewed distribution of unemployment rates (see Figure 1), alternative classification schemes such as equal intervals or quantiles would have obscured meaningful variation at the upper tail. Jenks classification therefore provides a more interpretable visual representation of spatial inequality in unemployment.

Spatial Autocorrelation and Clustering: Moran's I and LISA

Global Spatial Autocorrelation

To assess whether unemployment rates exhibited spatial clustering across Croydon, Global Moran's I was computed under the null hypothesis of spatial randomness. (Anselin,1995). A positive Moran's I statistic ($I \approx 0.02$) with a statistically significant z-score ($z = 9.62$, $p < 0.001$) indicated the presence of weak but significant positive spatial autocorrelation. This result confirms that neighbouring areas tend to exhibit similar unemployment levels, justifying further localised spatial analysis (Bivand & Wong, 2018).

Local Indicators of Spatial Association (LISA)

While global measures summarise overall clustering, they can mask important local variation. Local Moran's I (LISA) was therefore computed to identify statistically significant local clusters and spatial outliers.

This analysis revealed distinct High–High unemployment clusters concentrated in the northern parts of Croydon and Low–Low clusters in the south *(as shown in LISA cluster map presented in the main Findings section.)* Only statistically significant clusters ($p < 0.05$) were mapped to reduce the risk of spurious local inference (Bivand & Wong, 2018). These results provide actionable insights for geographically targeted employment interventions.

Neighbourhood Definition + Spatial Weights

Initial contiguity-based spatial weights matrices (Queen and Rook) produced unstable results due to irregular LSOA geometries at the borough periphery, leading to inconsistent neighbour counts and inflated Moran's I values outside the theoretical −1 to +1 range.

To address this, a distance-based spatial weights matrix was adopted. After testing multiple thresholds, a 2.5 km distance band was selected as it ensured full spatial connectivity while maintaining stable and interpretable autocorrelation statistics. Distance-based weights are recommended where irregular polygon geometries undermine contiguity-based neighbourhood definitions (Anselin, 2020).

Geographically Weighted Regression (GWR)

To examine whether the relationship between unemployment and deprivation varied spatially, Geographically Weighted Regression (GWR) was applied using IMD rank as an explanatory variable. GWR allows regression parameters to vary spatially, capturing non-stationary relationships that global models cannot detect (Brunsdon et al., 1996; Fotheringham et al., 2002).

An adaptive bandwidth was selected to account for uneven spatial density across Croydon, allowing the model to use smaller neighbourhoods in densely populated areas and larger neighbourhoods where data were sparser. This approach improves local parameter estimation compared to fixed-bandwidth alternatives (Comber, 2023).

Local Coefficients and Model Fit

To assess local model performance, local R^2 values were extracted from the GWR model and mapped (see Figure 6 in the Findings section). Local R^2 provides an indication of where deprivation explains unemployment variation more or less effectively across space.

Limitations

Several limitations should be noted. First, GWR does not produce reliable local p-values, and coefficient significance is approximated rather than directly tested(Fotheringham et al., 2002). Second, unemployment data reflect conditions during the COVID-19 pandemic period (ONS,2021), which may inflate rates in some areas relative to longer-term structural patterns.

Finally, all analyses are sensitive to the modifiable areal unit problem (MAUP) (Openshaw,1984) ; results may vary if alternative spatial units or aggregation schemes are applied.

Conclusion

The methodological choices employed in this analysis were selected to balance statistical robustness, interpretability, and policy relevance. Together, the combination of descriptive mapping, spatial autocorrelation analysis, and local regression provides a comprehensive framework for understanding both the distribution and spatial drivers of unemployment within Croydon.

Technical appendix references;

Lloyd, C.D., Norman, P.D. & McLennan, D. Deprivation in England, 1971–2020. *Appl. Spatial Analysis* **16**, 461–484 (2023). https://doi.org/10.1007/s12061-022-09486-8

Office for National Statistics (ONS) (2021) *Census 2021: Labour market and employment data*: https://www.ons.gov.uk

Anselin, L. (1995) *Local indicators of spatial association—LISA*. **Geographical Analysis**, 27(2), pp. 93–115.

Anselin, L. (2020) *Visualizing spatial autocorrelation*. GeoDa Workbook: https://geodacenter.github.io/workbook/5a_global_auto/lab5a.html

Brunsdon, C., Fotheringham, A.S. and Charlton, M. (1996) *Geographically weighted regression: a method for exploring spatial non-stationarity*. **Geographical Analysis**, 28(4), pp. 281–298.

Bivand, R. and Wong, D.W.S. (2018) *Comparing implementations of global and local indicators of spatial association*. **TEST**, 27(3), pp. 716–748.

Comber, A. (2023) *GEOG3915: Geocomputation and Spatial Analysis Practicals*: https://bookdown.org/lexcomber/GEOG3915/

Fotheringham, A.S., Brunsdon, C. and Charlton, M. (2002) *Geographically weighted regression: the analysis of spatially varying relationships*. Chichester: Wiley.

GISGeography (2023) *Choropleth maps – a guide to data classification*: https://gisgeography.com/choropleth-maps-data-classification/

Openshaw, S. (1984) *The modifiable areal unit problem*. Norwich: Geo Books.

Dataset description

Object name	Column IDs	Description	Object Source
Census 2021 Unemployment (OA-Level)	***OA21CD**	Output area code.	Derived from Office for National Statistics (ONS), Census 2021 dataset.
	***Unemployed**	Unemployment rate, originally measured at OA level and	

		aggregated to LSOA for analysis.	
English Indices of Deprivation 2025 (LSOA-level)	***IMD_rank-IMD rank (1=most deprived)** ***IMD_DECILE -IMD decile**	Composite deprivation index capturing multiple dimensions of socio-economic disadvantage at LSOA level.	The file used is... **‘**File_1_IoD2025_Index_of_Multiple_Deprivation.xlsx’ Ministry of Housing, Communities & Local Government (IMD 2025-October)
ONS LSOA 2021 Boundary Dataset	***LSOA21CD** ***geometry**	LSOA boundary polygons for England and Wales, used to spatially reference census + IMD attributes.	The file used is... ‘Lower_layer_Super_Output_Areas_2021_EW_BGC_V3.geojson’ Office for national statistics(ONS), Lower Layer Super Output Areas (December 2021) Boundaries.
Variables	**Description**		**Summary statistics : Mean(SD)**
Unemployed	Rate of economically active residents of Croydon who are unemployed within each LSOA, AGGREGATED FROM OA-Level Census 2021 data.		4.11 (1.40)
IMD_rank	Relative rank of deprivation for each LSOA in England, where lower values indicate higher deprivation.		15,587(9,227)
Spatial dataframes	**Description**		**Source**
Croydon_lsoa_data	LSOA-level spatial dataframe for Croydon, combining Census unemployment , IMD 2025 indicators and LSOA boundary geometries.		Derived (Joined Census 2021 data + IMD 2025 to ONS LSOA boundary files.
Croydon_lisa	Extended version of croydon_lsoa_data containing Local Moran’s I statistics, spatial lags, and LISA cluster classifications.		Derived(Local indicators of Spatial Association)

RSCRIPT

```
# OFFICIAL Rscript for spatial analysis report on Croydon,London

#load packages
library(sf)
library(dplyr)
library(stringr)
library(ggplot2)
library(classInt)
library(ggspatial)

# 1) Read LSOA polygons (mapping layer)
lsoa <- st_read("Lower_layer_Super_Output_Areas_2021_EW_BGC_V3.geojson", quiet = TRUE) %>%
  st_transform(27700)

croydon_lsoa <- lsoa %>%
  filter(str_detect(LSOA21NM, regex("^Croydon\\b", ignore_case = TRUE)))

# 2) Read OA polygons ( OA file ALREADY contains LSOA21CD...)
oa <- st_read("OA_2021_EW_BGC_V2.shp", quiet = TRUE)

# If CRS missing, set it (your OA file is BNG / EPSG:27700)
if (is.na(st_crs(oa))) st_crs(oa) <- 27700
oa <- st_transform(oa, 27700)

# 3) Subset Croydon OAs using OA's own LSOA name field
oa_croydon <- oa %>%
  filter(str_detect(LSOA21NM, regex("^Croydon\\b", ignore_case = TRUE)))

# 4) Join unemployment table (employment with columns OA, Unemployed)
unemp_oa <- employment %>% select(OA, Unemployed)

oa_croydon <- oa_croydon %>%
  mutate(OA = OA21CD) %>%
  left_join(unemp_oa, by = "OA")

# 5) Aggregate OA unemployment up to LSOA (RATE => mean)
unemp_lsoa <- oa_croydon %>%
  st_drop_geometry() %>%
  group_by(LSOA21CD) %>%
  summarise(Unemployed = mean(Unemployed, na.rm = TRUE), .groups = "drop")

# 6) Join to Croydon LSOA polygons (this is used as final mapping object)
croydon_lsoa_data <- croydon_lsoa %>%
  left_join(unemp_lsoa, by = "LSOA21CD")
```

```r
# 7) check:
sum(is.na(croydon_lsoa_data$Unemployed))

x <- croydon_lsoa_data$Unemployed
x_ok <- x[is.finite(x)]

k <- min(5, length(unique(x_ok)))  # how many classes we can actually make
brks <- classIntervals(x_ok, n = k, style = "quantile")$brks

croydon_lsoa_data$unemp_grp <- cut(
  croydon_lsoa_data$Unemployed,
  breaks = brks,
  include.lowest = TRUE
)

#visual
# Descriptive statistics ( histogram)
ggplot(croydon_lsoa_data, aes(Unemployed)) +
  geom_histogram(
    bins = 30,
    fill = "#b2182b",
    colour = "white"
  ) +
  labs(
    title = "Distribution of unemployment rates (LSOA)",
    x = "Unemployment rate (%)",
    y = "Number of LSOAs"
  ) +
  theme_minimal()

#Plotting distribution of OA unemployment aggregated to LSOA

ggplot(croydon_lsoa_data) +
  geom_sf(aes(fill = unemp_grp), colour = "white", linewidth = 0.15) +
  scale_fill_brewer(palette = "Reds", na.value = "grey90", name = "Unemployment\n(quintiles)")
  labs(
    title = "Unemployment rate (%) by LSOA – Croydon",
    subtitle = "OA unemployment aggregated to LSOA",
    caption = "Source: Census OA unemployment; ONS LSOA boundaries"
  ) +
  annotation_scale(location = "bl", width_hint = 0.35) +
  annotation_north_arrow(location = "tl", style = north_arrow_fancy_orienteering) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid.major = element_line(colour = "grey90", linewidth = 0.2),
    legend.position = "right"
```

```r
#PART B - Adding 2025 IMD
library(readxl)
library(dplyr)

imd <- read_excel("File_1_IoD2025_Index_of_Multiple_Deprivation.xlsx", sheet= 2 )

#inspect column names
names(imd)

#rename imd variables
imd <- imd %>%
  rename(
    LSOA21CD  = `LSOA code (2021)`,
    IMD_rank  = `Index of Multiple Deprivation (IMD) Rank (where 1 is most deprived)`,
    IMD_decile = `Index of Multiple Deprivation (IMD) Decile (where 1 is most deprived 10% of LSOA`
  )

#check
names(imd)
head(imd)

#Join imd onto Croydon LSOA map
croydon_lsoa_data <- croydon_lsoa_data %>%
  left_join(imd %>% select(LSOA21CD, IMD_rank, IMD_decile),
            by = "LSOA21CD")

#check
summary(croydon_lsoa_data$IMD_decile)
sum(is.na(croydon_lsoa_data$IMD_decile))
```

```r
# imd PLOT
library(ggplot2)
library(ggspatial)

reds10 <- c("#fff5f0","#fee0d2","#fcbba1","#fc9272","#fb6a4a",
            "#ef3b2c","#cb181d","#a50f15","#67000d","#3b0000")

ggplot(croydon_lsoa_data) +
  geom_sf(aes(fill = factor(IMD_decile)), colour = "white", linewidth = 0.15) +
  scale_fill_manual(
    values = reds10,
    name = "IMD decile\n(1 = most deprived)",
    na.value = "grey90"
  ) +
  labs(
    title = "Index of Multiple Deprivation (2025) by LSOA – Croydon",
    subtitle = "IMD deciles (1 = most deprived 10%)",
    caption = "Source: IMD 2025; ONS LSOA boundaries"
  ) +
  annotation_scale(location = "bl", width_hint = 0.35) +
  annotation_north_arrow(location = "tl", style = north_arrow_fancy_orienteering) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid.major = element_line(colour = "grey90", linewidth = 0.2),
    legend.position = "right"
  )

#PART C - spatial autocorrelation measures

#1) Global model (OLS)
ols <- lm(Unemployed ~ IMD_rank, data = croydon_lsoa_data)

summary(ols)

#2) Diagnosing residuals (MORAN'S I)
#spatial neighbours- contiguity
library(spdep)

# neighbours
nb <- poly2nb(croydon_lsoa_data, queen = TRUE)

# spatial weights
lw <- nb2listw(nb, style = "W", zero.policy = TRUE)

#Attach OLS residuals
```

```r
#Moran's I on residuals
moran_ols <- moran.test(
  croydon_lsoa_data$ols_resid,
  lw,
  zero.policy = TRUE
)

moran_ols

#LOCAL MORAN'S I- LISA cluster mapfor unemployment
library(spdep)
library(sf)
library(dplyr)

croydon_lisa <- st_make_valid(croydon_lsoa_data)

# neighbours + weights
nb_u <- poly2nb(croydon_lisa, queen = TRUE)
lw_u <- nb2listw(nb_u, style = "W", zero.policy = TRUE)

# local Moran's I
x <- croydon_lisa$Unemployed
lm_u <- localmoran(x, lw_u, zero.policy = TRUE)

#check col names
colnames(lm_u)

croydon_lisa <- croydon_lisa %>%
  mutate(
    lisa_I = lm_u[, "Ii"],
    lisa_p = lm_u[, "Pr(z != E(Ii))"],
    x_z = as.numeric(scale(x)),
    lag_x_z = as.numeric(lag.listw(lw_u, x_z, zero.policy = TRUE))
  )
```

```r
alpha <- 0.05
croydon_lisa <- croydon_lisa %>%
  mutate(
    lisa_cluster = case_when(
      lisa_p > alpha ~ "Not significant",
      x_z >= 0 & lag_x_z >= 0 ~ "High-High",
      x_z <= 0 & lag_x_z <= 0 ~ "Low-Low",
      x_z >= 0 & lag_x_z <= 0 ~ "High-Low",
      x_z <= 0 & lag_x_z >= 0 ~ "Low-High",
      TRUE ~ "Not significant"
    ),
    lisa_cluster = factor(
      lisa_cluster,
      levels = c("High-High","Low-Low","High-Low","Low-High","Not significant")
    )
  )

#LISA MAP
ggplot(croydon_lisa) +
  geom_sf(aes(fill = lisa_cluster), colour = "white", linewidth = 0.15) +
  scale_fill_manual(
    values = c(
      "High-High" = "#b2182b",
      "Low-Low" = "#2166ac",
      "High-Low" = "#ef8a62",
      "Low-High" = "#67a9cf",
      "Not significant" = "grey92"
    ),
    name = "LISA cluster type\n(p < 0.05)"
  ) +
  annotation_scale(location = "bl", width_hint = 0.35) +
  annotation_north_arrow(location = "tl", style = north_arrow_fancy_orienteering) +
  labs(
    title = "Spatial clustering of unemployment (LISA) — Croydon",
    subtitle = "High–High / Low–Low clusters and spatial outliers",
    caption = "Source: Census OA unemployment aggregated to LSOA; Local Moran's I"
  ) +
  theme_minimal(base_size = 12) +
  theme(panel.grid.major = element_line(colour = "grey90", linewidth = 0.2))


#4) Local Model(GWR)
library(spgwr)
library(sf)

# ensure projected CRS
croydon_gwr <- st_transform(croydon_lsoa_data, 27700)
```

```r
# extract coordinates
coords <- st_coordinates(st_centroid(croydon_gwr))

#Bandwidth selection
bw <- gwr.sel(
Unemployed ~ IMD_rank,
data = as(croydon_gwr, "Spatial"),
coords = coords,
adapt = TRUE
)

bw

#Run GWR
gwr_model <- gwr(
Unemployed ~ IMD_rank,
data = as(croydon_gwr, "Spatial"),
coords = coords,
adapt = bw,
hatmatrix = TRUE,
se.fit = TRUE
)

#Attach outputs back to sf
croydon_gwr$IMD_coef <- gwr_model$SDF$IMD_rank
croydon_gwr$local_R2 <- gwr_model$SDF$localR2

summary(croydon_gwr$IMD_coef)
summary(croydon_gwr$local_R2)

#Mapping GWR outputs
#a) local IMD coefficients
ggplot(croydon_gwr) +
geom_sf(aes(fill = IMD_coef)) +
scale_fill_viridis_c(
option = "plasma",
name = "Local IMD coefficient"
) +
labs(
title = "Spatial variation in IMD-unemployment relationship",
subtitle = "GWR local coefficients"
) +
theme_minimal()
```